MATHEMATICAL OLYMPIAD PREPARATION

From an Idea to a Proof

A practical guide to turning a promising observation into a complete mathematical argument.

Proof laboratory 01 Read, try, explain, improve

How to use this guide. Read a short section at a time. You can attempt an exercise independently, ask for a worked explanation, or return later. The printable exercises omit hints and answers; the HTML contains both. You may stop, ask questions, return later or work without a timer. This is a learning resource, not an examination.

START HERE

Read the problem before choosing a theorem

Mathematics begins with a precise task. A familiar-looking question can hide a new condition.

01
GIVEN

What objects are allowed? Look for positive, integer, distinct, for every, on a segment, on a line, adjacent, and nonzero.

02
TARGET

What must the last line establish: an existence claim, the exact maximum, an impossibility, all solutions, or equality?

03
GAP

What has already been shown, and what is only suspected? Describe what a counterexample would look like.

WORKED EXAMPLE

Finding answers is not the same as verifying them

Find all real numbers x such that $\sqrt{(x + 2)} = x$.

1. **Domain:** The square root is nonnegative, so any solution has $x \geq 0$.
2. **Candidate step:** Squaring gives $x + 2 = x^2$, hence $(x - 2)(x + 1) = 0$. The candidates are $x = 2$ and $x = -1$.
3. **Check the original:** $x = 2$ gives $\sqrt{4} = 2$. The candidate -1 is invalid, since $\sqrt{1} \neq -1$.

The only solution is $x = 2$. Squaring supplied necessary candidates; checking finished the job.

Your four-line starting template

Given: …
To prove: …
Already forced: …
Still to establish: …

These four lines frame a proof. They do not replace it.

STUDY NOTE

The proof finish line

The most important difference is between seeing a pattern and showing why it must be true.

A good idea is the beginning of a proof, not the end. Before starting, identify what the question is asking you to establish. Most olympiad problems require more than a numerical answer.

01
Claim

Record the permitted objects and the exact target. For a maximum or minimum, a bound and an attainable example have different jobs. For a classification, show all solutions have the proposed form and check the surviving candidates.

02
Reason

Link your observations by justified steps. No unexplained leaps.

03
Close

State exactly why the conclusion follows. Check every case.

Four different proof tasks

A construction shows one example can work. A universal bound shows every allowed example obeys a limit. A necessary condition holds for all actual solutions but may allow extras; a sufficient condition guarantees all original requirements. Classifications need both directions or justified equivalences.

WORKED MODEL

A complete existence and optimality proof

1. Problem: two nonnegative real numbers have sum 12. What is the largest possible product?
2. **Upper bound:** from $(a - b)^2 \geq 0$, we have $(a + b)^2 \geq 4ab$.
3. As $a + b = 12$, we obtain $144 \geq 4ab$ and therefore $ab \leq 36$.
4. **Construction:** choose $a = b = 6$. Then $ab = 36$.

Finish the proof: The product cannot exceed 36, and the choice (6, 6) achieves 36. Hence the maximum is exactly 36.

Closing checklist

What have I proved? Does the proof cover *every* allowed case? If I guessed a formula, have I proved both that no other answers exist and that mine work? If I claim a maximum, have I given both the bound and a construction?

PRACTICAL TOOLKIT

Eight methods to reach for first

A technique is a move. Choosing it for a reason is a method. Explaining every necessary step is a proof.

Do not memorise thirty theorem names today. Learn the eight common moves below. Each card tells you what job it can finish and what it cannot do.

01 **Geometry correspondence**

02 **Candidates and verification**

03 **Factorisation and squares**

04 **Induction**

05 **Residues and parity**

06 **Bound and construction**

07 **Pigeonhole**

08 **Invariants**

> **What about geometry?**
>
> Geometry is taught separately because corresponding vertices, directed angles, and auxiliary lines need careful diagrams. Start with the worked similarity proof below. The full extended toolkit includes geometry tools.

STUDY NOTE

Algebra

Make each transformation lawful. Separate possible answers from verified answers.

In algebra, the crucial habit is to make each line follow from the last. A correct identity can be useful without yet proving the target statement.

Check the direction of your reasoning

$$(a - b)^2 \geq 0$$
$$\Rightarrow a^2 + b^2 \geq 2ab$$

This establishes the inequality for every real a and b. Equality occurs precisely when a = b.

Two tests before writing the next line

Degree: under a scaling a → ta, b → tb, a homogeneous degree-3 expression scales by t^3. If an identity is homogeneous, scaling all variables by t multiplies every term of degree 3 by t^3. Degree matching catches some mistakes, but not incorrect coefficients. Mixed degrees can be legitimate when a given constraint or substitution explains them. **Equivalence:** if you square, divide or cancel an expression, explain why the step is valid, especially if a term might be zero or negative.

WORKED MODEL

Solving a functional equation on the integers

1. Suppose $g : \mathbb{Z} \to \mathbb{Z}$ satisfies $g(x + y) = g(x) + g(y)$ for all integers x, y. Find every such function.
2. Putting $x = y = 0$ gives $g(0) = g(0) + g(0)$, hence $g(0) = 0$.
3. Let $c = g(1)$. Setting $y = 1$ gives $g(n + 1) = g(n) + c$. At $n = 0$, $g(0) = 0 = c \cdot 0$. If $g(n) = cn$, the recurrence gives $g(n + 1) = cn + c = c(n + 1)$. Thus induction proves $g(n) = cn$ for all nonnegative integers n.
4. Putting $y = -x$ gives $g(-x) = -g(x)$. Hence $g(n) = cn$ for negative integers too.
5. Conversely, for any integer c define $g(n) = cn$. For every integer pair x, y, $g(x + y) = c(x + y) = cx + cy = g(x) + g(y)$. Thus all such candidates work, so this is the complete classification.

Finish the proof: Every solution has the form $g(n) = cn$ with $c \in \mathbb{Z}$, and every one of these functions works.

> **When using induction**
>
> State the starting value, the induction hypothesis and the exact statement for the next value. In a recurrence, copy coefficients carefully. For example, an equation with 2f(1) must not silently become f(1).

METHOD SELECTION

AM-GM, Schur, and the direction of a bound

A true inequality can still be insufficient. The question is not only whether a tool is valid, but whether its conclusion moves the proof towards the target.

WORKED EXAMPLE

First diagnose the proposed route

For positive a, b, c, prove

$$(a + b + c)^3 + 9abc \geq 4(a + b + c)(ab + bc + ca).$$

Write $p = a + b + c$, $q = ab + bc + ca$ and $r = abc$. The target becomes $p^3 + 9r \geq 4pq$.

AM-GM correctly gives $p^3 \geq 27r$, and therefore $p^3 + 9r \geq 36r$. But to conclude the target via this route we would also need $36r \geq 4pq$. At $(a, b, c) = (1, 1, 2)$, the bound gives 72 while the target right side is 80. So that bridge does not follow.

What went wrong? This particular AM-GM step provides too weak a lower bound for the entire left side. It does not show that AM-GM is incapable of proving the inequality.

PROOF 1

Expand, order, and factor

Set D equal to the left side minus the right side. Direct expansion gives

$$D = a^3 + b^3 + c^3 + 3abc - (a^2b + a^2c + b^2a + b^2c + c^2a + c^2b).$$

This is symmetric, so relabel $a \geq b \geq c > 0$. The identity below follows by expansion, not guesswork:

$$D = (a - b)^2(a + b - c) + c(a - c)(b - c).$$

Both terms are nonnegative: the square is nonnegative; $a + b - c$ is positive; and c, $a - c$, $b - c$ are nonnegative. Thus $D \geq 0$.

If $D = 0$, the first term forces $a = b$, and the second then forces $a = c$ since $c > 0$. Conversely all-equal variables give equality. Hence equality holds exactly when $a = b = c$.

Schur's inequality

The inequality is often called Schur's cubic inequality. The name is useful later, but it is not a prerequisite for understanding either proof. For a first encounter, explore an expansion, the equality case, and one hint about factoring.

Direction checklist: What quantity is bounded? Is the bound an upper or lower bound? Is that the direction your target needs? Do the equality cases fit? If the proof uses division, do you know the sign?

PROOF COMPLETION

What induction proves, and what checking proves

Induction can establish a candidate family perfectly. A complete classification must also ensure every retained candidate satisfies every original condition.

WORKED EXAMPLE

When an extra condition eliminates candidates

Find all functions $g : \mathbb{Z} \to \mathbb{Z}$ satisfying, for every pair of integers x, y, both

$$g(x + y) = g(x) + g(y) \quad \text{and} \quad g(xy) = g(x)g(y).$$

1. **Find candidates:** Additivity forces $g(0) = 0$. With $c = g(1)$, the recurrence $g(n + 1) = g(n) + c$ and induction give $g(n) = cn$ for $n \geq 0$. Since $g(-n) = -g(n)$, this extends to all integers.
2. **Check the other requirement:** The multiplicative equation at $x = y = 1$ gives $c = c^2$. Therefore $c = 0$ or $c = 1$. In particular the additive candidate $g(n) = 2n$ fails.
3. **Verify the survivors:** The zero function and identity function both satisfy additivity and multiplicativity for every integer pair.

Thus exactly $g(n) = 0$ and $g(n) = n$ work. A general recurrence alone did not decide which constants survive the second condition.

For the original quadratic equation $f(x+y) + f(x-y) = 2f(x) + 2f(y)$, the proposed family $f(n) = cn^2$ passes the universal check because

$$c(x+y)^2 + c(x-y)^2 = 2cx^2 + 2cy^2.$$

By contrast, $f(n) = cn^2 + k$ gives $2cx^2 + 2cy^2 + 2k$ on the left but $2cx^2 + 2cy^2 + 4k$ on the right. It satisfies the equation only when $k = 0$.

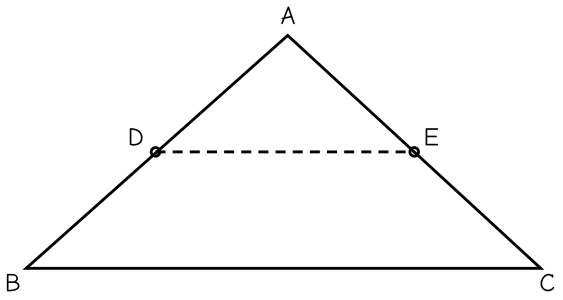
> **Nuance:**
>
> If every step of your proof is already reversible, the final check may be built into that argument. What matters is proving both logical directions, not mechanically repeating a ritual.

STUDY NOTE

Geometry

Use the diagram to discover; use named relationships to prove.

A diagram is a place to discover relationships. Your written proof must then explain why those relationships hold. A picture that looks convincing is not a justification on its own.



A labelled diagram helps to map equal ratios and corresponding angles.

WORKED MODEL

A midpoint proof

1. Let D and E be the midpoints of AB and AC respectively. We want to show DE ∥ BC and DE = BC/2.
2. By the midpoint conditions, AD/AB = AE/AC = 1/2.
3. The included angle ∠DAE equals ∠BAC because D lies on AB and E lies on AC.
4. Therefore triangles ADE and ABC are similar by the side-angle-side similarity criterion.
5. Corresponding angles are equal, giving ∠ADE = ∠ABC; hence DE ∥ BC. Also DE/BC = AD/AB = 1/2.

Finish the proof: Both the parallelism and the length relation follow from the same justified similarity statement.

Geometry proof order

State which points are collinear or perpendicular. Name the theorem you use: corresponding angles, cyclic quadrilaterals, similarity, or the tangent-chord theorem. State the vertex correspondence before using similarity. For triangles ADE and ABC it is A to A, D to B, E to C, so DE corresponds to BC. The labels must match the proven angles and side ratios. If feet lie on extensions, ordinary angle arguments may need directed angles.

STUDY NOTE

Number theory

Residues, exact divisibility and the logic of infinite descent.

Trying small values is excellent for discovery. To finish, explain why the pattern must hold for all relevant integers.

Modular arithmetic as a filter

Modulo 5, the inputs 0, 1, 2, 3, 4 have square residues 0, 1, 4, 4, 1 respectively. This complete mapping explains which residue can yield a square divisible by 5. This gives a short proof of an important fact.

WORKED MODEL

A divisibility proof

1. Suppose 5 divides n^2.
2. Any integer is congruent to 0, 1, 2, 3 or 4 modulo 5. The corresponding square residues are 0, 1, 4, 4, 1, in that order.
3. The only way n^2 can be congruent to 0 modulo 5 is when $n \equiv 0 \pmod 5$.

Finish the proof: Thus 5 divides n. The argument covers every integer, not just tested examples.

Infinite descent

Descent is a contradiction strategy. Suppose a nonzero integer solution exists. Choose one with a *positive integer measure* as small as possible. Show that its existence would force another solution with a strictly smaller positive measure. This is impossible.

WORKED MODEL

A miniature descent

1. Assume $x^2 = 2y^2$ has a nonzero integer solution. If $y = 0$, the equation gives $x = 0$, contrary to the nonzero assumption. So $y \neq 0$, and among nonzero solutions choose one with the smallest positive $|y|$.
2. x^2 is even, so x is even. Write $x = 2u$. Then $y^2 = 2u^2$, so y is also even; write $y = 2v$.
3. Substitution gives $u^2 = 2v^2$, a new integer solution.
4. But $0 < |v| = |y|/2 < |y|$, contradicting minimality.

Finish the proof: There is no nonzero integer solution to $x^2 = 2y^2$.

> **Exact divisibility versus divisibility**
>
> For nonzero N, to show $v_3(N) = r$ you need **both** $3^r \mid N$ and $3^{r+1} \nmid N$. Showing only that 3^r divides N gives a lower bound, not the exact exponent.

STUDY NOTE

Combinatorics

Find what is preserved, what is unavoidable and what makes a bound sharp.

Combinatorics frequently rewards finding a quantity that stays unchanged, a property that must always hold, or a construction showing a bound is sharp.

Find an invariant

If a process repeatedly changes a configuration, ask: what expression could stay the same after every legal move?

WORKED MODEL

A multiplicative invariant

1. Start with the entries 1, 2, 3. A move replaces any two entries a, b with a + b + 2ab.
2. Observe that 1 + 2(a + b + 2ab) = (1 + 2a)(1 + 2b).
3. Therefore the product of all factors (1 + 2 × entry) does not change after a move.
4. Initially it is 3 × 5 × 7 = 105. If the last entry is T, then 1 + 2T = 105.

Finish the proof: T = 52, regardless of which legal pair is combined first.

Universal claims need universal arguments

An example proves that a configuration **can** exist. To prove that every configuration has a feature, you need a theorem or argument that covers all configurations, not just the one drawn on your page.

WORKED MODEL

Sharpness using the pigeonhole principle

1. Problem: find the least N such that any N integers contain two whose difference is divisible by 5.
2. **Upper bound:** there are only five possible remainders modulo 5. Among six integers, two share a remainder.
3. Subtracting those two integers gives a multiple of 5.
4. **Counterexample for smaller N:** the five integers 0, 1, 2, 3, 4 all have distinct remainders.

Finish the proof: Six works for every choice; five can fail. Therefore the least N is 6.

STUDY NOTE

Your practice laboratory

Short exercises with optional hints and model proofs.

These are **guided learning questions**, not scored examination items. Attempt just two or three in a sitting. Each has a hint and a complete model argument you can open when you are ready. Try to give a reason for every step.

01 **ALGEBRA**

Prove that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for every integer $n \geq 1$.

Focus: Use every part of induction.

02 **ALGEBRA**

For real numbers x and y, prove $x^2 + y^2 \geq 2xy$ and give the equality condition.

Focus: Give both the bound and equality case.

03 **ALGEBRA**

Let $h : \mathbb{Z} \to \mathbb{Z}$ satisfy $h(x + y) = h(x) + h(y)$ **for all integers x and y**. Prove $h(-n) = -h(n)$ for every integer n.

Focus: Use the original quantifier correctly.

04 **GEOMETRY**

In triangle ABC, D and E are the midpoints of AB and AC. Prove DE = BC/2.

Focus: Name the triangles and prove similarity.

05 **GEOMETRY**

A triangle has interior angles α, β and γ. Prove $\alpha + \beta + \gamma = 180°$.

Focus: Do not use the picture alone as justification.

06 **NUMBER THEORY**

Prove that if 5 divides n^2, then 5 divides n.

Focus: Explain why the modular possibilities are exhaustive.

07 **NUMBER THEORY**

Show that 7 divides $8^n - 1$ for every integer $n \geq 1$.

Focus: Move from examples to all n.

08 **NUMBER THEORY**

Explain why a proof of $v_3(N) = 2$ needs more than proving 9 divides N. Give an example.

Focus: Distinguish exact from at least.

09 **COMBINATORICS**

Six integers are chosen. Prove that two have a difference divisible by 5.

Focus: Make the universal argument explicit.

10 **COMBINATORICS**

Starting with 2, 3, 4, repeatedly replace any pair a, b by $a + b + 2ab$. Prove that the final number is 157 regardless of move order.

Focus: Specify the invariant and why it is preserved.

11 **PROOF STRUCTURE**

A student finds an example reaching a claimed maximum. What additional step is needed to prove it really is the maximum?

Focus: Tell existence apart from optimality.

12 **PROOF STRUCTURE**

Let $g : \mathbb{Z} \to \mathbb{Z}$ satisfy both $g(x + y) = g(x) + g(y)$ and $g(xy) = g(x)g(y)$ for all integers x, y. The first condition forces $g(n) = cn$ for some integer c. Determine which c work and prove the complete answer.

Focus: Separate candidate discovery from a full list of solutions.

UNSEEN-STYLE PRACTICE

Transfer laboratory

Six new exercises. Their titles do not tell you exactly which move will finish them.

Start by writing Given, Target, Already forced, and Still to prove. These are learning exercises with optional hints, not an official BMO paper. If you open a hint, keep that attempt classified as guided practice.

F1

ALGEBRA / INEQUALITY

For real $t > 0$, prove $t + 1/t \geq 2$, with equality only when $t = 1$. What changes for $t < 0$?

Transfer exercise: choose your own tool before checking the hint.

F2

NUMBER THEORY / RESIDUES

Prove that no integers x, y satisfy $x^2 + y^2 \equiv 3 \pmod 4$.

Transfer exercise: choose your own tool before checking the hint.

I1

ALGEBRA / FUNCTIONAL EQUATIONS

Find all $f : \mathbb{Z} \to \mathbb{Z}$ with $f(x + y) + f(x - y) = 2f(x)$ for all integers x, y.

Transfer exercise: choose your own tool before checking the hint.

I2

COMBINATORICS / INVARIANTS

Three lamps are either on or off. One move flips exactly two lamps. Initially only the first lamp is on. Prove all-off is impossible, but all-on is reachable.

Transfer exercise: choose your own tool before checking the hint.

C1

ALGEBRA / SHARP BOUNDS

Positive x, y, z satisfy $x + y + z = 3$. Prove $xy + yz + zx \leq 3$. State equality.

Transfer exercise: choose your own tool before checking the hint.

C2

COMBINATORICS / COUNTEREXAMPLE

Does a positive integer N exist such that EVERY set of N distinct integers contains two whose sum is divisible by 3? Justify your answer.

Transfer exercise: choose your own tool before checking the hint.

SUGGESTED NEXT STEPS

Three gentle learning sessions

Begin with tools that connect directly to your current questions. Choose whether to use hints.

Session 1, 20 to 40 minutes: Read 'What induction proves', then try I1. Explain in words why a candidate list is not always a full classification.

Session 2, 20 to 40 minutes: Read the AM-GM lesson, do F1 and, if useful, C1. Ask whether a bound points in the needed direction.

Session 3, 20 to 40 minutes: Practise I2 and C2. Notice what an invariant proves and how a counterexample disproves a universal claim.

Sessions may be split, skipped, repeated or changed. Short written arguments matter more than completing the schedule.

STUDY NOTE

A gentle seven-session plan

A flexible way to learn without making every meeting an examination.

This is a suggested order, not a requirement to finish within seven consecutive days. A session can take about 20 to 40 minutes. Stop earlier if tired.

Session 1: Proof finish line; exercises 11 and 12.

Session 2: Algebra; exercises 1, 2 and 3.

Session 3: Geometry; exercises 4 and 5.

Session 4: Number theory; exercises 6, 7 and 8.

Session 5: Combinatorics; exercises 9 and 10.

Session 6: Choose two earlier proofs. Rewrite them more clearly without opening the model.

Session 7: Review what felt natural, what required hints and which techniques you want to revisit.

A proof-writing template

Claim: I need to show that…

Given: The conditions allow me to assume…

Key idea: The useful observation is…

Justification: This implies the next step because…

Conclusion: Therefore the required statement follows…

Final check: I have included all cases, checked equality or sufficiency where needed and verified the arithmetic.

> **A useful stopping rule**
>
> If an approach stops producing useful facts, write what it established and what remains. You can take a break, request one hint, or switch representation for a stated reason. A timer is optional and says nothing about your ability. A thoughtful partial argument is still valuable learning.